

Toyota 89650-02B50 EPS ECU Hardware Mod Guide

Trace cutting, tap-point soldering, and panda mounting (for Prius C)

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⚠ Safety Warning

EPS (Electric Power Steering) is a vehicle active-safety system. This mod involves cutting PCB traces and precision soldering; a wrong cut or a short can cause loss of steering. A modified ECU must not be used in an occupied vehicle on public roads. Perform all testing in a controlled environment. You assume full responsibility for any damage or accident resulting from this modification.

1. Materials & Tools

1.1 Main parts

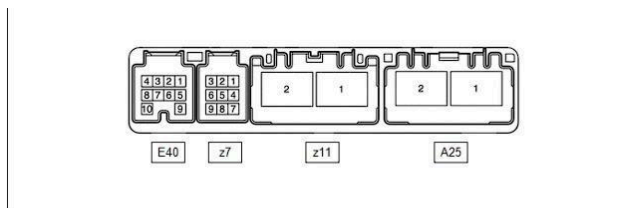
Item	Description	Notes
EPS ECU	Toyota 89650-02B50 (JTEKT / DENSO, EPS 12V)	Gas-model unit
panda	3-channel CAN interface board	Acts as the gateway
Wiring	6 signal wires: CAN0 H/L, CAN2 H/L, 12V, GND AWG26~28 stranded, twist the CAN pairs	Photos use red / green
Mounting	Velcro / zip tie to fix the panda onto the ECU case	See FIG4

1.2 Tools

- Soldering iron (fine tip, temp-controlled preferred), solder, flux
- Hobby knife / scalpel (to cut PCB traces), tweezers
- Multimeter (verify traces are cut, check CAN-pair continuity, measure 12V/GND)
- Magnifier / microscope (tap points are tiny — strongly recommended)

2. Pinout & Voltage Reference (FIG0)

The table below is the factory connector pinout, wire colors, and measurement specs for the 89650-02B50. Use it to identify the CAN traces to cut and the power pins, and to verify by measurement before and after the mod.



*a	Component without harness connected (Power Steering ECU Assembly)	-	-
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(a) Measure the voltage and resistance according to the value(s) in the table below.

NOTICE:

- Check if the power steering ECU assembly connectors are securely connected by pulling on them from the backside.
- If the EPS warning light is illuminated due to a malfunction, the fail-safe function may cause the voltage of the power steering ECU's terminals to become 0 V.

TERMINALS NO. (SYMBOL)	WIRING COLOR	TERMINAL DESCRIPTION	CONDITION	SPECIFIED CONDITION
E40-1 (IG) - Body ground	L - Body ground	IG power source voltage	Ignition switch ON	8 to 16 V
E40-6 (TS) - Body ground	SB - Body ground	Test mode signal	Always	9 to 16 V
E40-7 (CANH) - E40-8 (CANL)	GR - W	CAN communication line	Ignition switch off	54 to 69 Ω
z7-1 (TRQ2) - z7-2 (TRQG)	Y - B	Torque sensor 2 input signal	Engine running, steering wheel not being turned (without load)	2.3 to 2.7 V
			Engine running, steering wheel being turned to right	1.2 to 2.5 V
			Engine running, steering wheel being turned to left	2.5 to 3.8 V
z7-2 (TRQG) - Body ground	B - Body ground	Torque sensor ground	Always	Below 1 Ω
z7-8 (TRQV) - z7-2 (TRQG)	R - B	Torque sensor power source	Ignition switch ON	4.5 to 5.5 V
z7-9 (TRQ1) - z7-2 (TRQG)	W - B	Torque sensor 1 input signal	Engine running, steering wheel not being turned (without load)	2.3 to 2.7 V
			Engine running, steering wheel being turned to right	2.5 to 3.8 V
			Engine running, steering wheel being turned to left	1.2 to 2.5 V
z11-1 (M1) - Body ground	R - Body ground	Power steering motor assembly output signal 1	Engine running, steering wheel being turned to right	9 to 16 V
			Engine running, steering wheel being turned to left	6 to 8 V
z11-2 (M2) - Body ground	B - Body ground	Power steering motor assembly output signal 2	Engine running, steering wheel being turned to right	6 to 8 V
			Engine running, steering wheel being turned to left	9 to 16 V
A25-1 (PIG) - Body ground	L - Body ground	Power steering motor assembly power supply input signal	Always	9 to 16 V
A25-2 (PGND) - Body ground	W-B - Body ground	Power ground	Always	Below 1 Ω

FIG0 — 89650-02B50 factory terminal table (E40 / z7 / z11 / A25 pinout and measurement specs)

The pins most relevant to this mod:

Terminal	Wire color	Signal
E40-1 (IG)	L	IG power (8~16 V with ignition ON)
E40-7 (CANH)	GR	CAN High — CUT this trace
E40-8 (CANL)	W	CAN Low — CUT this trace
A25-1 (PIG)	L	Motor power input (always 9~16 V)
A25-2 (PGND)	W-B	Power ground (always <1 Ω)

Between E40-7 and E40-8, the factory spec with ignition OFF is 54~69 Ω (the bus termination resistor). After cutting the traces the chip side and connector side are separated, so this value changes — use it to confirm the traces are truly cut.

3. Hardware Modification Steps

Core idea: cut the two CAN traces E40-7 / E40-8 so the connector side and the EPS chip side are separated, then splice the panda in between — CAN0 to the connector side, CAN2 to the chip side, with 12V/GND tapped from the board.

Step 1: Disassemble the ECU and locate the work area

1. Remove the ECU case screws, take out the PCB, and flip to the solder side (back).
2. Find the area at the bottom-left, near the connector — this is the trace-cutting and soldering work area (red box in FIG1).

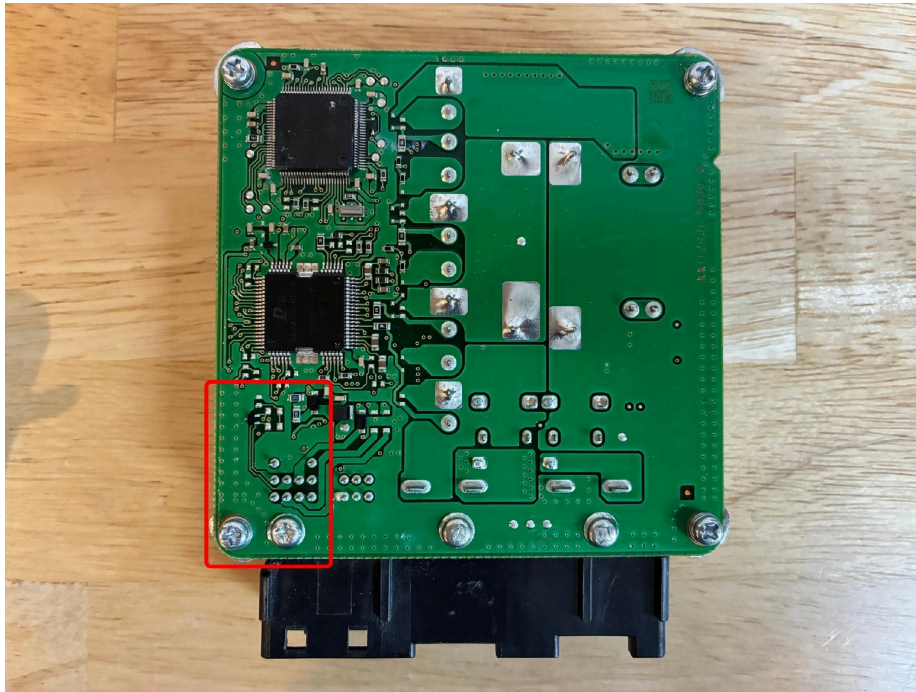


FIG1 — PCB back side. Red box marks the work area (bottom-left, near the black connector)

Step 2: Cut the two traces E40-7 / E40-8 (CUT)

The most critical and irreversible step. Before cutting, use a multimeter and cross-check FIG0 and FIG2 to confirm these are exactly the E40-7 (CANH) / E40-8 (CANL) traces going to the chip.

3. At the blue ⊗ **CUT** mark in FIG2, locate the two traces for E40-7 / E40-8.
4. Score through both traces with the knife; after scraping, confirm no continuity across each cut with the meter to ensure they are fully open.

Step 3: Solder the 6 tap points

Following FIG2, find the six pads and solder the signal wires. FIG2 (bottom-left) also shows the pin order on the panda side.

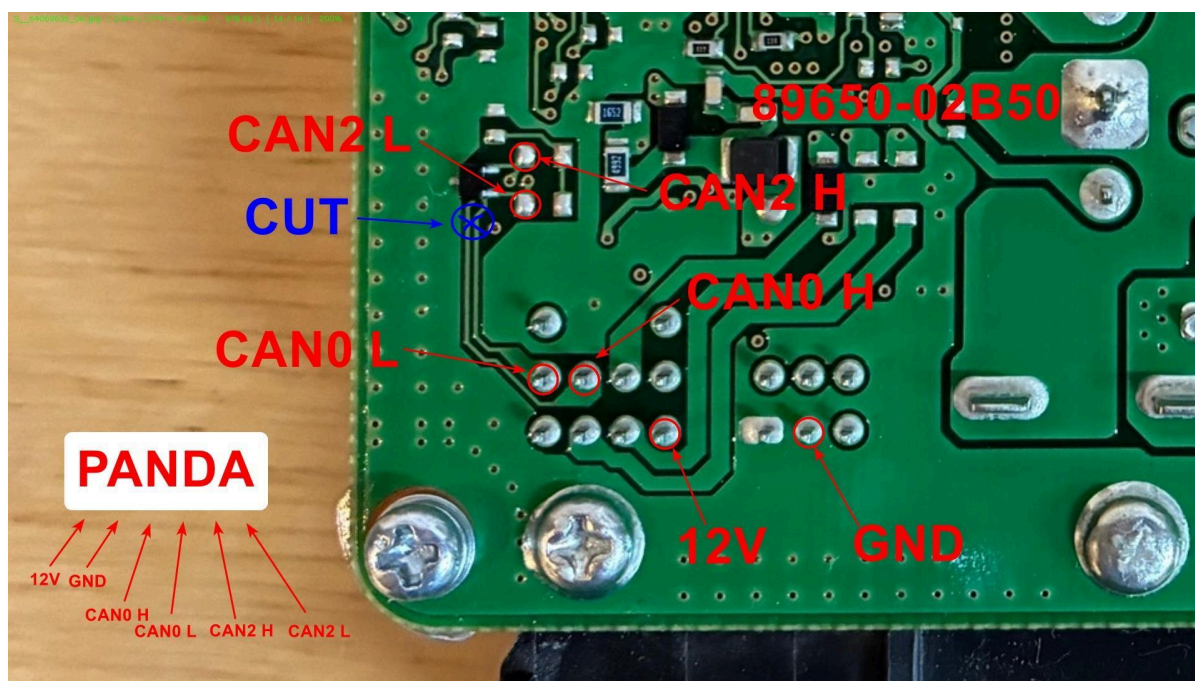


FIG2 — tap-point map: CAN2 H/L, CAN0 H/L, 12V, GND, plus the blue CUT point; bottom-left shows the PANDA pin order

Pad	Description	To panda
CAN0 H	Connector-side CAN High (E40-7 side)	CAN0 H
CAN0 L	Connector-side CAN Low (E40-8 side)	CAN0 L
CAN2 H	Chip-side CAN High	CAN2 H
CAN2 L	Chip-side CAN Low	CAN2 L
12V	Board power tap (positive)	12V
GND	Ground	GND

PANDA pin order (FIG2 bottom-left, left to right):

12V GND CAN0 H CAN0 L CAN2 H CAN2 L

CAN is a differential signal — twist each H/L pair, and run 12V/GND as another pair. The pads are small; work under magnification and avoid bridging adjacent pads. Before applying power, verify each wire has continuity to its panda pin and that adjacent pins are not shorted.

Step 4: Wire to the panda

Connect the six wires to the panda per Step 3. FIG3 shows the finished soldering; the board is also hand-labeled in marker (L, 12V, GND, and the 02B50 part reference) to aid future servicing.

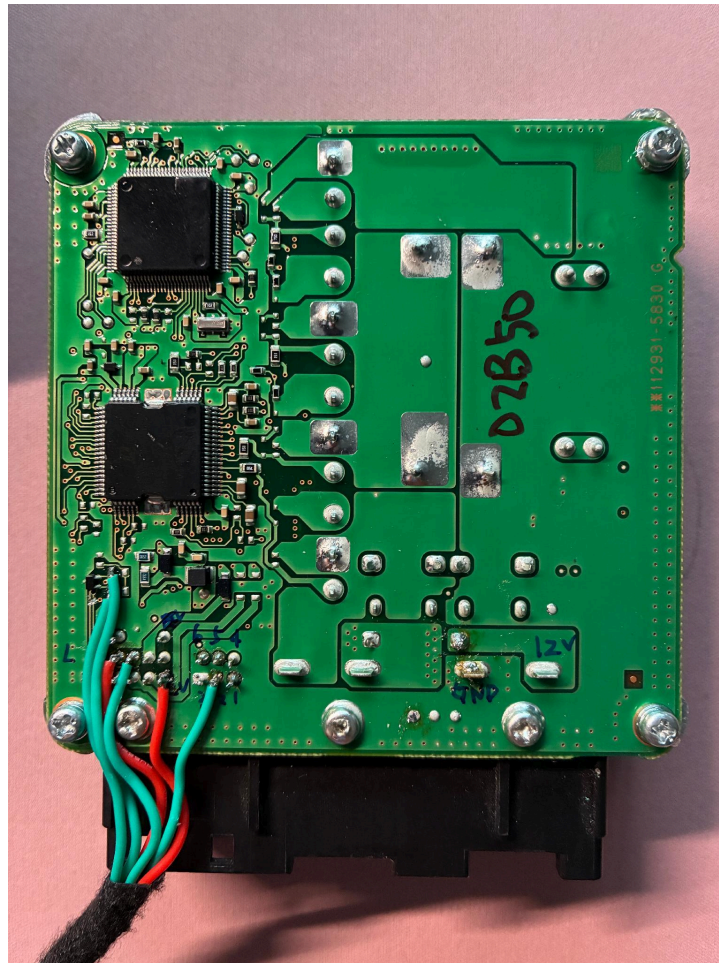


FIG3 — six wires soldered, pads hand-labeled on the board

Step 5: Mount the panda and reassemble

Fix the panda onto the EPS ECU case with velcro / a zip tie (FIG4) so it becomes a self-contained module. When routing wires, keep them from being pinched by the case or pulling on the solder joints.

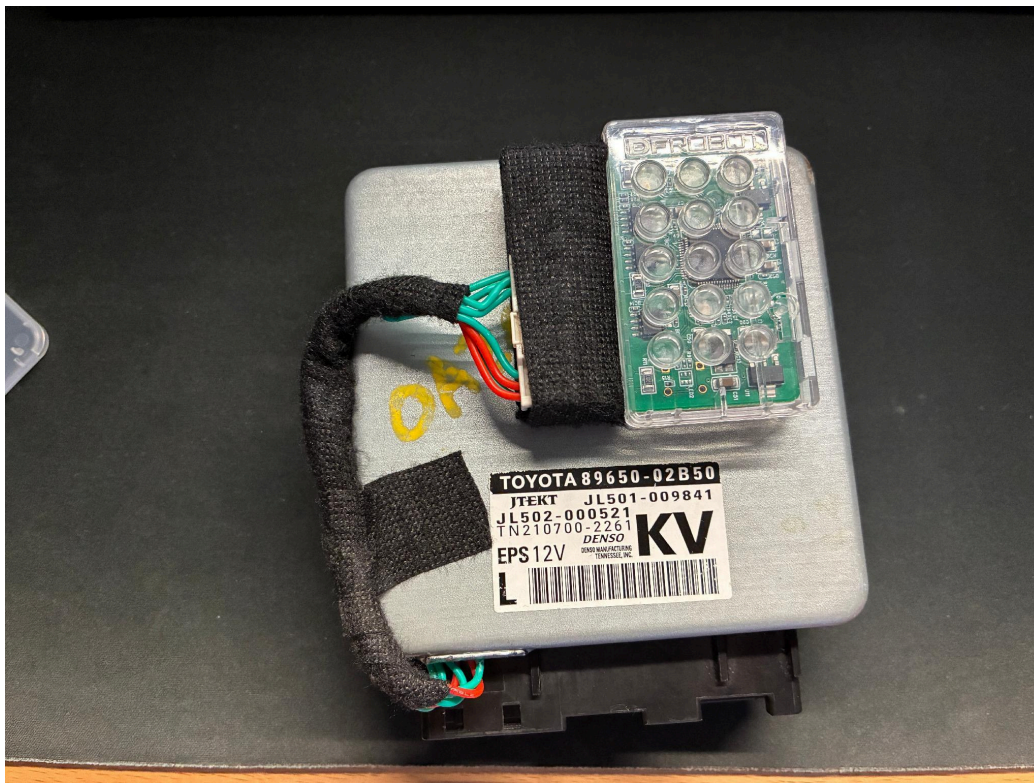


FIG4 — finished unit: panda mounted on the 89650-02B50 (TOYOTA / JTEKT / DENSO, EPS 12V)

4. Pre-Power Hardware Checks

1. Traces E40-7 / E40-8 are cut (no continuity across each).
2. Each of the 6 wires has continuity to its corresponding panda pin.
3. No shorts between adjacent pads (especially the CAN pairs and 12V/GND).
4. No short between 12V and GND.
5. Only after all of the above check out, apply power.

Appendix: Wiring Quick Reference

panda pin	Pad (EPS PCB)	Meaning
12V	Board 12V tap	panda power (positive)
GND	Board GND	Ground
CAN0 H	E40-7 (CANH) side tap	Connector-side CAN High
CAN0 L	E40-8 (CANL) side tap	Connector-side CAN Low
CAN2 H	Chip-side tap	EPS chip CAN High
CAN2 L	Chip-side tap	EPS chip CAN Low
—	E40-7 / E40-8 trace	CUT — separates connector from chip